

Multiple Choice (10 marks)

1. Which of the following does authentication aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is
2. A proxy firewall filters at the
 - (a) Physical layer
 - (b) Application layer
 - (c) Data link layer
 - (d) Network layer
3. In Message Confidentiality, the transmitted message must make sense to only intended
 - (a) Receiver
 - (b) Sender
 - (c) Modulator
 - (d) Translator
4. Can we build a secure PRP from a secure PRG?
 - (a) No, it cannot be done
 - (b) Yes, just plug the GGM PRF into the Luby-Rackoff theorem
 - (c) It depends on the underlying PRG
 - (d) Option text
5. Which of the following is not a security exploit?
 - (a) Eavesdropping
 - (b) Cross-site scripting
 - (c) Authentication
 - (d) SQL Injection

True / False (10 marks)

Answer True or False

6. True or False: Wireshark is a widely used tool for capturing and analyzing wireless traffic in network security.
7. True or False: A buffer overflow can allow direct writing to the instruction pointer register, which executes the attacker's code immediately.
8. True or False: A ping sweep is used to identify live systems on a network by sending ICMP echo requests to multiple IP addresses.
9. True or False: Message authentication is a service that ensures message integrity by verifying the authenticity of the message sender.
10. True or False: Snort is a tool used for network intrusion detection and real-time traffic analysis.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to calculate the sum of series $1^3+2^3+3^3+\dots+n^3$.
12. Write a function to insert a given string at the beginning of all items in a list.

End of Paper